

AI based research paper summarization system

ADT

LAB Exercise – 2

1. Create the necessary tables for the identified database objects from the ER Diagram of your business domain with the necessary constraints (Primary Key, Unique, Not Null, Check, Default, and Foreign Key).

```
1 • CREATE DATABASE paper_summarization;
```

```
1 • CREATE TABLE User (  
2     UserID INT PRIMARY KEY,  
3     FullName VARCHAR(100) NOT NULL,  
4     Email VARCHAR(100) NOT NULL UNIQUE,  
5     Role VARCHAR(20) NOT NULL,  
6     ProfilePicture VARCHAR(255),  
7     DateJoined DATE DEFAULT (CURRENT_DATE),  
8     CHECK (Role IN ('Student', 'Researcher'))  
9 );  
10  
--
```

```
1 • CREATE TABLE Student (  
2     StudentID INT PRIMARY KEY,  
3     department VARCHAR(100) NOT NULL,  
4     year_of_study int(10) NOT NULL,  
5     FOREIGN KEY (StudentID)  
6         REFERENCES User(UserID)  
7         ON DELETE CASCADE  
8         ON UPDATE CASCADE  
9 );  
10
```

```
1 • ○ CREATE TABLE Researcher (  
2     ResearcherID INT PRIMARY KEY,  
3     bio VARCHAR(100) NOT NULL,  
4     ResearchArea VARCHAR(100) NOT NULL,  
5     FOREIGN KEY (ResearcherID)  
6         REFERENCES User(UserID)  
7         ON DELETE CASCADE  
8         ON UPDATE CASCADE  
9 );  
10
```

```
1 • ○ CREATE TABLE ResearchPaper (  
2     PaperID INT PRIMARY KEY,  
3     Title VARCHAR(200) NOT NULL,  
4     Author VARCHAR(150) NOT NULL,  
5     sourcee VARCHAR(100) NOT NULL,  
6     pages int(10) NOT NULL,  
7     UserID INT NOT NULL,  
8     UploadDate DATE DEFAULT (CURRENT_DATE),  
9     FOREIGN KEY (UserID)  
10        REFERENCES User(UserID)  
11        ON DELETE CASCADE  
12        ON UPDATE CASCADE  
13 );  
14
```

```
1 • ○ CREATE TABLE AIModel (  
2     ModelID INT PRIMARY KEY,  
3     ModelName VARCHAR(100) NOT NULL,  
4     Version VARCHAR(20) NOT NULL,  
5     Accuracy DECIMAL(5,2) NOT NULL,  
6     CHECK (Accuracy BETWEEN 0 AND 100)  
7 );  
8
```

```
1 • ○ CREATE TABLE Summary (  
2     SummaryID INT PRIMARY KEY,  
3     SummaryText TEXT NOT NULL,  
4     SummaryLength INT NOT NULL,  
5     PaperID INT NOT NULL,  
6     ModelID INT NOT NULL,  
7     CHECK (SummaryLength > 0),  
8     FOREIGN KEY (PaperID)  
9         REFERENCES ResearchPaper(PaperID)  
10        ON DELETE CASCADE  
11        ON UPDATE CASCADE,  
12    FOREIGN KEY (ModelID)  
13        REFERENCES AIModel(ModelID)  
14        ON DELETE CASCADE  
15        ON UPDATE CASCADE  
16 );
```

```
• ○ CREATE TABLE SummaryHistory (  
    HistoryID INT PRIMARY KEY,  
    UserID INT NOT NULL,  
    SummaryID INT NOT NULL,  
    Notes VARCHAR(255),  
  
    FOREIGN KEY (UserID)  
        REFERENCES User(UserID)  
        ON DELETE CASCADE  
        ON UPDATE CASCADE,  
  
    FOREIGN KEY (SummaryID)  
        REFERENCES Summary(SummaryID)  
        ON DELETE CASCADE  
        ON UPDATE CASCADE  
);
```

```

CREATE TABLE Feedback (
    FeedbackID INT PRIMARY KEY,
    Rating INT NOT NULL,
    Commentt VARCHAR(255),
    UserID INT NOT NULL,
    PaperID INT NOT NULL,

    CHECK (Rating BETWEEN 1 AND 5),

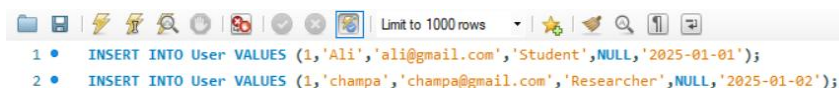
    FOREIGN KEY (UserID)
        REFERENCES User(UserID)
        ON DELETE CASCADE
        ON UPDATE CASCADE,

    FOREIGN KEY (PaperID)
        REFERENCES ResearchPaper(PaperID)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);

```

2. Demonstrate the enforcement of constraints with necessary inputs.

1. PRIMARY KEY Constraint



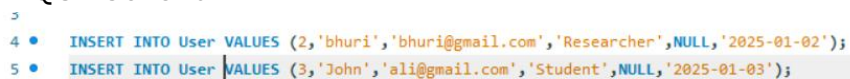
```

1 • INSERT INTO User VALUES (1, 'Ali', 'ali@gmail.com', 'Student', NULL, '2025-01-01');
2 • INSERT INTO User VALUES (1, 'champa', 'champa@gmail.com', 'Researcher', NULL, '2025-01-02');

```

Error: Duplicate entry for primary key.

2. UNIQUE Constraint



```

3
4 • INSERT INTO User VALUES (2, 'bhuri', 'bhuri@gmail.com', 'Researcher', NULL, '2025-01-02');
5 • INSERT INTO User VALUES (3, 'John', 'ali@gmail.com', 'Student', NULL, '2025-01-03');

```

Error: Duplicate email.

3. NOT NULL Constraint



```

6
7 • INSERT INTO User VALUES (4, 'rock@gmail.com', 'student', NULL, '2026-01-04');
8 • INSERT INTO User VALUES (5, 'Tom', 'tom@gmail.com', 'student', NULL, '2026-01-04');

```

Error: FullName cannot be NULL.

4. CHECK Constraint

```
10 • INSERT INTO User VALUES (6,'jack','jack@gmail.com','student',NULL,'2026-01-04');
11 • INSERT INTO User VALUES (7,'bro','bro@gmail.com','teacher',NULL,'2026-01-04');
```

Error: CHECK constraint violated.

5. FOREIGN KEY Constraint

```
13 • INSERT INTO User VALUES (8,'Nimal','nimal@gmail.com','Student',NULL,'2026-03-05');
14 • INSERT INTO ResearchPaper VALUES (102,'ML Research','David','ML',15,1,'2025-01-06');
15
16 -- here id = 20 is not exist in User table , therefore it gave an error
17 • INSERT INTO ResearchPaper VALUES (103,'AI Research','john','AI',20,'2025-01-06');
```

Error: User ID 20 does not exist in User table.

6. DEFAULT Constraint

```
19 -- INSERT INTO User (UserID, FullName, Email, Role, ProfilePicture) VALUES (11, 'Kamal', 'kamal@gmail.com', 'Student', null);
20 • SELECT * FROM User
21 WHERE UserID = 11;
```

Result Grid						
Filter Rows:						
Edit: Export/Import: Wrap Cell Content:						
UserID	FullName	Email	Role	ProfilePicture	DateJoined	
11	Kamal	kamal@gmail.com	Student	NULL	2026-06-14	
•	NULL	NULL	NULL	NULL	NULL	

Result: DateJoined is automatically assigned.

3.Demonstrate the enforcement of constraints using the ALTER TABLE command.

```
22
23 • ALTER TABLE AIModel MODIFY ModelName VARCHAR(100) NOT NULL;
```

4. List the constraints enforced on each table.

```
1  SELECT
2      TABLE_NAME,
3      CONSTRAINT_NAME,
4      CONSTRAINT_TYPE
5  FROM INFORMATION_SCHEMA.TABLE_CONSTRAINTS
6  WHERE TABLE_SCHEMA = DATABASE();
```

TABLE_NAME	CONSTRAINT_NAME	CONSTRAINT_TYPE
aimodel	PRIMARY	PRIMARY KEY
aimodel	aimodel_chk_1	CHECK
feedback	PRIMARY	PRIMARY KEY
feedback	feedback_ibfk_1	FOREIGN KEY
feedback	feedback_ibfk_2	FOREIGN KEY
feedback	feedback_chk_1	CHECK
researchpaper	PRIMARY	PRIMARY KEY
researchpaper	researchpaper_ibfk_1	FOREIGN KEY
student	PRIMARY	PRIMARY KEY
student	student_ibfk_1	FOREIGN KEY
summary	PRIMARY	PRIMARY KEY
summary	summary_ibfk_1	FOREIGN KEY
summary	summary_ibfk_2	FOREIGN KEY
summary	summary_chk_1	CHECK
summaryhistory	PRIMARY	PRIMARY KEY
summaryhistory	summaryhistory_ibfk_1	FOREIGN KEY
summaryhistory	summaryhistory_ibfk_2	FOREIGN KEY
user	PRIMARY	PRIMARY KEY
user	user_chk_1	CHECK

5. Demonstrate dropping of any one constraint of an object.

```
24
25 • ALTER TABLE User DROP INDEX Email;
26
```

6. Demonstrate how data integrity is maintained by propagating changes across parent and child tables present in the database.

Demonstration 1: ON UPDATE CASCADE.

```
27 • INSERT INTO User VALUES (12,'bhumil','bhumil@gmail.com','Student',NULL,ProfilePicture );
28 • INSERT INTO ResearchPaper VALUES (103,'AI Research','John Smith','IEEE',2,17," 2026-10-05");
29 • UPDATE User
30 SET UserID = 10
31 WHERE UserID = 1;
```

7. Demonstrate the Drop and Truncate Commands on the objects and display the records.

```
32
33 • TRUNCATE TABLE Researcher;
34
```

```
DROP TABLE Researcher;
```

```
32
33 • SELECT * FROM Researcher;
34
```

Table 'paper_summerization.researcher' doesn't exist.